String Slicing in Python

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Python slicing is about obtaining a sub-string from the given string by slicing it respectively from start to end.

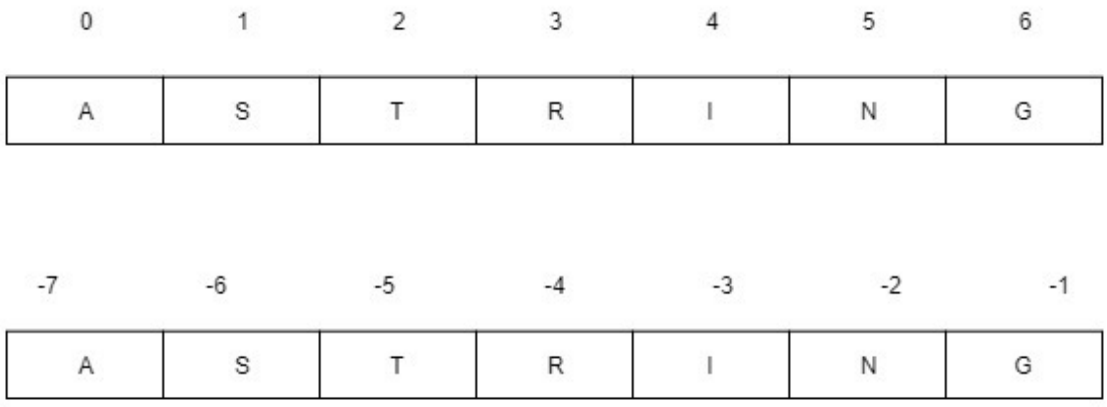
How String slicing in Python works

For **understanding slicing** we will use different methods, here we will cover 2 methods of string slicing, one using the in-build slice() method and another using the **[:] array slice**. String slicing in Python is about obtaining a sub-string from the given string by slicing it respectively from start to end.

Python slicing can be done in two ways:

- Using a slice() method
- Using the array slicing [::] method

Index tracker for positive and negative index: String indexing and slicing in python. Here, the Negative comes into consideration when tracking the string in reverse.



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Method 1: Using the slice() method

The slice() constructor creates a slice object representing the set of indices specified by range(start, stop, step).

Syntax:

- *slice(stop)*
- *slice(start, stop, step)*

Parameters: start: *Starting index where the slicing of object starts.* ***stop:*** *Ending index where the slicing of object stops.* ***step:*** *It is an optional argument that determines the increment between each index for slicing.* ***Return Type:*** *Returns a sliced object containing elements in the given range only.*

Example:

Python3

```
# Python program to demonstrate
# string slicing

# String slicing
String = 'ASTRING'

# Using slice constructor
```

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```
print("String slicing")
print(String[s1])
print(String[s2])
print(String[s3])
```

Output:

```
String slicing
AST
SR
GITA
```

Method 2: Using the List/array slicing [::] method

In Python, indexing syntax can be used as a substitute for the slice object. This is an easy and convenient way to slice a string using list slicing and Array slicing both syntax-wise and execution-wise. A start, end, and step have the same mechanism as the slice() constructor.

Below we will see **string slicing in Python with examples.**

Syntax

```
arr[start:stop]         # items start through stop-1
arr[start:]             # items start through the rest of the
array
arr[:stop]              # items from the beginning through stop-
1
arr[:]                  # a copy of the whole array
arr[start:stop:step]    # start through not past stop, by step
```

Example 1:

In this example, we will see **slicing in python list** the index start from 0 indexes and ending with a 2 index(stops at 3-1=2).

Python3

```
# Python program to demonstrate
```

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```
# String slicing
String = 'GEEKSFORGEEKS'

# Using indexing sequence
print(String[:3])
```

Output:

Example 2:

In this example, we will see the example of starting from 1 index and ending with a 5 index(stops at 3-1=2), and the skipping step is 2. It is a good example of **Python slicing string by character.**

Python3

```
# Python program to demonstrate
# string slicing

# String slicing
String = 'GEEKSFORGEEKS'

# Using indexing sequence
print(String[1:5:2])
```

Output:

```
EK
```

Example 3:

In this example, we will see the example of starting from -1 indexes and ending with a -12 index(stops at 3-1=2)and the skipping step is -2.

Python3

```
# Python program to demonstrate
# string slicing
```

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```
String = 'GEEKSFORGEEKS'

# Using indexing sequence
print(String[-1:-12:-2])
```

Output:

```
SEGOSE
```

Example 4:

In this example, the whole string is printed in reverse order.

Python3

```
# Python program to demonstrate
# string slicing

# String slicing
String = 'GEEKSFORGEEKS'

# Prints string in reverse
print(String[::-1])
```

Output:

```
SKEEGROFSKEEG
```

Note: To know more about strings click here.

Using islice()

The islice() is a built-in function defined in itertools module. It is used to get an iterator which is an index-based slicing of any iterable. It works like a standard slice but returns an iterator.

Syntax:

itertools.islice(iterable, start, stop[, step])
Parameters: iterable: Any iterable sequence like list, string, tuple etc. start: The start index from where the slicing of iterable starts. stop: The end index

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specifies the gap between each index for slicing. Return Type: Return an iterator from the given iterable sequence.

Example:

Python3

```
# Python program to demonstrate
# islice()

import itertools

# Using islice()
String = 'GEEKSFORGEEKS'

# prints characters from 3 to 7 skipping one character.
print(''.join(itertools.islice(String, 3, 7)))
#This code is contributed by Edula Vinay Kumar Reddy
```

Output

```
KSF0
```

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Last Updated : 27 Mar, 2023

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